

PAPER NO. 2102 - SELECTED QUESTIONS

ORDINARY DIFFERENTIAL EQUATIONS

UNIT I - LINEAR DIFFERENTIAL EQUATIONS

4. The given $y_1=e^x$, $y_2=e^{2x}$ and $y_3=e^{3x}$ are linearly independent solutions of the third order homogeneous equation $y''' - 6y'' + 11y' - 6y = 0$. Find a particular solution of the given equation that satisfies $y(0)=0$, $y'(0)=0$, $y''(0)=3$.

6. Determine the appropriate form for a particular solution of $y'' + 6y' + 13y = e^{-3x} \cos 2x$.

8. Solve the initial value problem $y'' - 2y' + 2y = x + 1$; $y(0)=3$, $y'(0)=0$.

9. Find a particular solution of the given differential equation $y'' + 3y' + 2y = 4e^x$ by using the method of variation of parameters.

UNIT II - POWER-SERIES SOLUTIONS

5. Solve the equation $(x - 3)y' + 2y = 0$ by using power series method. Determine the radius of convergence of the resulting series.

6. Solve the equation $(x - 1)y' + 2y = 0$ by using power series method.

7. Solve the initial value problem $x^2 + 1)y'' + 2xy' - 2y = 0$; $y(0)=0$, $y'(0)=1$ by using power series method.

ASSIGNMENT 1

1. (i) Verify that y_1 and y_2 are solutions of the differential equation $y'' - 2y' + 2y = 0$; $y_1=e^x \cos x$, $y_2=e^x \sin x$.

(ii) Find the general solution of $9y'' - 12y' + 4y = 0$.

4. Solve the initial value problem $y''' + 3y'' - 10y' = 0$; $y(0)=7$, $y'(0)=0$, $y''(0)=70$.

ASSIGNMENT 2

4. Find a power series solution of the differential equation $y'' + y = 0$.